Beyond behavioural fit: Reconstructing belief dynamics reveals cognitive bottlenecks and adaptive mechanisms in human flexible learning

Abstract

Inferring internal cognitive processes from behavior is an ill-posed inverse problem: distinct latent beliefs and strategies can generate the same observable choices. Here we introduce a modular Bayesian modeling architecture for trial-resolved inference of each individual's latent cognitive states, jointly capturing two sources of individual differences: cognitive resources that limit information availability and inductive biases that shape evidence interpretation and strategy sampling.

Applied to novel category learning in a four-dimensional feature space under three rules of increasing complexity, our model and several alternative deep neural-network models all accurately predicted individual choices; only our model, however, faithfully reconstructed each learner's evolving feature-use strategies, independently validated against trial-by-trial verbal reports never used for model fitting.

Analysis of the reconstructed dynamics revealed how the interplay of hypothesis search, belief updating, and belief-to-choice mapping gave rise to distinct learning profiles. Rapid learners focused search on simpler hypotheses and sustained belief accumulation despite occasional errors, whereas slow learners searched broadly and repeatedly reset beliefs, compensating for limited memory but delaying stabilization. Interestingly, the model traced "aha" moments to the sudden access to prior learning rather than sudden rule discovery, suggesting that useful beliefs accumulated outside awareness before becoming consciously available to guide choice reliably.

Together, these findings provide a mechanistic account of why individuals learn differently, laying a computational foundation for individualized learning interventions and the design of human-inspired AI that learns flexibly under resource constraints.

A category choice can be supported by different internal explanations. Two learners may give the same answer while attending to different features, considering different rules or drawing on different amounts of accumulated evidence. Understanding individual learning therefore requires an account of how these internal processes develop

across trials. The central problem is to infer their dynamics from an observable sequence of stimuli, choices and feedback.

The model distinguishes three aspects of a learner's strategy: the rules currently considered, the evidence supporting each rule and each rule's influence on choice. These aspects can change at different times. A useful rule may become available and accumulate support while a previously used rule continues to guide responses. This distinction allows learning to be examined in terms of rule discovery, evidence retention and the translation of belief into behavior.

We formulated a modular Bayesian architecture that connects these processes within a single learning trajectory. Capacity, memory and perceptual uncertainty constrain the information available to the learner; search preferences and persistence in rule use shape how that information affects subsequent choices. Fitting the model to choices yields a distribution of possible internal trajectories for each individual. These trajectories provide the basis for independent strategy validation and for examining how cognitive resources and adaptive mechanisms contribute to different learning profiles.

Results

Category learning and individual learning trajectories

The task required learning to classify stimuli defined by four feature lengths. Feedback distinguished correct and incorrect category responses; in the hierarchical condition, partial credit additionally identified a response from the correct family but the wrong species. These observations provide a trial-by-trial record of the information available to the learner and the choices made from it. We used this setting to formulate a common account of how candidate rules are discovered, evaluated and expressed in behavior across feedback conditions.

[Behavioral results to be added.]

A modular model separates hypothesis availability, belief and choice

The model represents learning as a changing set of explanations for the category structure. A rule specifies how stimulus features determine a category, for example through a feature threshold or a comparison between features. Three quantities distinguish its role in learning. Membership in the candidate set determines whether a rule is currently considered. Rule belief measures its relative support from feedback. Choice weight determines how strongly it influences the current response. A finite capacity limits the number of candidates, while search and abandonment allow their identities and number to change across trials.

Five cognitive modules implement this learning process: perception (P), evidence memory (M), hypothesis updating (H), precision updating (B) and choice readout (R). P supplies the perceived stimulus to the active rules; R combines their category predictions into a choice distribution. After feedback, M updates rule beliefs, B adjusts prediction precision and H prepares the candidate set for the next trial.

Before feedback arrives, the available rules jointly generate a choice. Perception (P) introduces uncertainty into the feature values supplied to each rule. A rule then assigns probabilities to categories, with its precision controlling how concentrated these predictions are. Choice readout (R) combines the predictions according to rule beliefs while giving an advantage to the previous trial's dominant rule, which had the largest choice weight. This inertia permits persistent rule use while evidence for alternatives accumulates. As beliefs change, choice weights can spread across several rules or concentrate on one rule within the same learner (see Methods).

Feedback changes the relative support for every available rule according to how well it predicts the observed outcome. Evidence memory (M) determines how much earlier support is retained when new feedback is incorporated. Precision updating (B) calibrates how sharply each rule predicts categories. The precision gradient indicates whether increasing or decreasing precision makes the observed feedback more likely under that rule. A single learning rate controls adjustment in both directions. Belief updating measures relative support among rules, while precision updating adjusts the concentration of each rule's category predictions. In the hierarchical condition, feedback also updates beliefs about which response categories belong to the same family. This knowledge carries forward as the candidate rules change.

Hypothesis updating (H) regulates access to alternative explanations. Recent failure can increase search and favor a broader range of candidates. Local search modifies an existing rule or considers a rule with similar classification outcomes; global search draws from the wider rule catalogue. Rules with weak support are more likely to leave the candidate set. A newly considered rule inherits a share of existing support according to its classification agreement with retained rules. Subsequent feedback then updates its belief and precision alongside the other candidates.

Together, these processes distinguish the discovery of a rule from the accumulation of belief and its expression in choice. Their timing can diverge: an increase in belief can precede a change in the rule that dominates responses. Perceptual uncertainty, candidate capacity and evidence retention constrain what information is available; search preferences and rule-use inertia shape its use. The resulting trajectories connect observable learning profiles to specific points at which information is acquired, retained or translated into behavior.

The learner's internal trajectory remains uncertain. For each participant, particle inference tracks multiple possible trajectories and evaluates how well their predicted choices match the observed sequence. Participant parameters remain fixed while the internal states evolve. The inferred states are distributions over supported explanations of behavior, providing a basis for subsequent validation and mechanistic analysis (see Methods).

Choice prediction and model comparison

[Results to be added.]

Independent validation of reconstructed strategies

[Results to be added.]

Cognitive resources and adaptive mechanisms across learning profiles

[Results to be added.]

Belief-to-choice transitions during rapid learning improvements

[Results to be added.]

Recovery and robustness of inferred dynamics

[Results to be added.]

Discussion

[Discussion to be completed after the empirical results.]

Methods

Task and latent learning state

Each trial presents a stimulus $\boldsymbol{x}_t \in [0,1]^4$, followed by a category response y_t and feedback r_t. The four coordinates encode normalized feature lengths. Condition 1 has two categories and binary feedback; condition 2 has four categories and binary feedback. Condition 3 has four categories grouped into two families. Its feedback distinguishes the correct species ($r_t = 1$), the correct family but incorrect species ($r_t = 0.5$), and the incorrect family ($r_t = 0$). Participants in this condition know the family–species structure and learn the response-label pairings.

The model represents a classification rule as a partition of stimulus space into labeled category regions. A finite catalogue $\mathcal{H}$ contains rules based on feature thresholds, comparisons and conditional judgments. The learner considers an active subset $\mathcal{A}_t$, with size bounded by the participant's capacity $M^{\max}$. Each active rule h has a normalized belief $\pi_t^-(h)$ and a prediction precision $\beta_t(h)$. The learner starts with one uniformly sampled rule, belief 1 and initial precision β_0. Superscripts $-$ and $+$ denote beliefs before choice and after feedback. Participant indices are omitted; parameters remain fixed within a participant while these states evolve.

Perception and choice precede feedback on each trial. Feedback then updates beliefs and precisions, after which hypothesis updating prepares the candidate set for the next trial. Thus, changes induced by current feedback influence subsequent choices.

[Participant recruitment, task administration, exclusions and ethics information to be added.]

Perception (P): representing stimulus uncertainty

Perception (P) introduces feature-specific uncertainty calibrated from independent perceptual measurements. All active rules receive the same perceived stimulus $\widetilde{\boldsymbol{x}}_t$

within a learning trajectory. A rule predicts categories according to the distance $d_{t,h,c}$ from this stimulus to its region for category c:

$$p_{t,h}(c) = \frac{\exp[-\beta_t(h) d_{t,h,c}]}{\sum_{a=1}^{C} \exp[-\beta_t(h) d_{t,h,a}]}, \tag{1}$$

where C is the number of categories. Distance is the shortest Euclidean distance to the category region, and equals zero inside it. Greater precision makes category probabilities more sensitive to these distance differences.

Choice readout (R): mapping beliefs to choices

Choice readout (R) converts rule beliefs into contributions to the response. It also represents persistence in rule use: the previously dominant rule receives an advantage controlled by inertia λ. The choice weights and resulting category probabilities are

$$w_t(h) = \frac{\pi_t^-(h) \exp[\lambda \mathbb{I}(h = h_{t-1}^\star)]}{\sum_{v \in \mathcal{A}_t} \pi_t^-(v) \exp[\lambda \mathbb{I}(v = h_{t-1}^\star)]}, \tag{2}$$

$$P_t(c \mid z_t) = \sum_{h \in \mathcal{A}_t} w_t(h) p_{t,h}(c), \tag{3}$$

where $\mathbb{I}$ is the indicator function and z_t denotes the internal state, including the perceived stimulus. The current dominant rule $h_t^\star$ has the largest choice weight and supplies inertia on the next trial. When there is no preceding active dominant rule, weights equal beliefs. Distributed weights allow several rules to influence choice; concentrated weights allow one rule to dominate. Inertia can therefore preserve an established policy while evidence for alternatives accumulates.

Evidence memory (M): updating rule beliefs

Evidence memory (M) evaluates how well every active rule predicts the feedback received after the chosen response. For binary feedback, the likelihood $\ell_t(h)$ and updated belief are

$$\ell_t(h) = r_t p_{t,h}(y_t) + (1 - r_t)[1 - p_{t,h}(y_t)], \tag{4}$$

$$\pi_t^+(h) = \frac{[\pi_t^-(h)]^\gamma \ell_t(h)}{\sum_{v \in \mathcal{A}_t} [\pi_t^-(v)]^\gamma \ell_t(v)}. \tag{5}$$

Correct feedback supports probability assigned to the chosen category; incorrect feedback supports the remaining categories. The retention parameter $0 \leq \gamma \leq 1$ controls how much previous evidence persists. Values near 1 preserve accumulated belief differences, while smaller values reduce these differences before new evidence enters. Normalization produces relative support within the current candidate set. All active candidates undergo this update, including those contributing little to choice.

Precision updating (B): adjusting prediction precision

Precision updating (B) calibrates the concentration of each rule's category predictions using the observed feedback. Let $\ell_t(h;\beta)$ denote the feedback likelihood evaluated at precision β, using Eq. (4) for binary feedback and the pairing-averaged likelihood below for condition 3. Its log-likelihood gradient determines the direction of adjustment:

$$\kappa_t(h) = \left.\frac{\partial \log \ell_t(h;\beta)}{\partial \beta}\right|_{\beta=\beta_t(h)}. \tag{6}$$

The derivative holds category distances and pre-feedback pairing beliefs fixed. Positive gradients favor sharper predictions; negative gradients favor broader predictions, according to their effect on the feedback likelihood.

A single participant-level learning rate $\eta \in [0,1]$ controls the update in both directions:

$$\beta_{t+1}(h) = \mathrm{clip}\big(\beta_t(h) + \eta\kappa_t(h),\, \beta_{\min},\, \beta_{\max}\big). \tag{7}$$

Clipping restricts precision to the shared bounds $\beta_{\min} = 0.1$ and $\beta_{\max} = 25$. All active rules update from their pre-feedback predictions, and the new precisions affect the next trial. The two participant-level parameters are initial precision β_0 and learning rate η.

Hypothesis updating (H): managing available rules

Hypothesis updating (H) controls which rules are available for learning. Search responds to recent performance through an accumulated failure state,

$$F_{t+1} = \delta F_t + (1-\delta)(1-r_t), \qquad F_1 = 0. \tag{8}$$

Failure retention $0 \le \delta < 1$ sets the persistence of this performance signal. Accumulated failure can increase the probability of searching, e_{t+1}, and the probability of a global source conditional on search, g_{t+1}:

$$e_{t+1} = \mathrm{sigmoid}[\mathrm{logit}(e_0) + a_e F_{t+1}], \tag{9}$$

$$g_{t+1} = g_0 + (1-g_0)a_g F_{t+1}. \tag{10}$$

Here e_0 and g_0 are baseline probabilities, a_e and a_g determine failure sensitivity, $\mathrm{sigmoid}(v) = 1/(1+e^{-v})$, and $\mathrm{logit}(p) = \log[p/(1-p)]$. Successful feedback reduces F, returning search toward its baseline.

After feedback learning, each rule has exit probability $\rho[1-\pi_t^+(h)]$, where ρ controls the overall exit rate. Weakly supported candidates are therefore more likely to leave, while at least one candidate remains available for prediction and learning.

Each search event introduces one rule. Local search selects a retained rule according to its belief and generates a related candidate through either a structural edit or functional similarity. Structural edits change one part of a rule, such as the feature used for a judgment. Edit distance measures the cost of this change, and candidates with smaller distances receive greater probability. Functional similarity measures the

proportion of reference stimuli assigned the same category by two rules. Candidates with greater agreement receive greater probability. A participant-level mixture weight controls these two local routes. Global search samples uniformly from the catalogue entries outside the current active set. The local and global routes allow hypotheses to change through focused revision and broader exploration.

If a search event would exceed capacity, the least-supported retained rule is removed. Candidate count therefore varies between one and $M^{\max}$. Retained beliefs are normalized before a share is transferred to the entrant. Transfer depends on functional affinity, which increases with classification agreement. Each old rule divides its belief between itself and the entrant; more strongly supported and more similar rules contribute larger amounts. This allocation preserves a total belief of one. Retained rules preserve their updated precisions, while entrants begin at β_0. This transfer allows a newly considered rule to build on existing classification knowledge.

Learning hierarchical category relations

Graded feedback provides evidence about both classification and family membership. In condition 3, the model maintains joint beliefs $\Pi_t(h, k)$ over rules and the three possible pairings k of four response labels. Pairing beliefs initially assign equal probability to each possibility. Let $\text{mate}_k(y)$ denote the other response in the same family under pairing k. The feedback likelihood is

$$\ell_t(h,k) = \begin{cases} p_{t,h}(y_t), & r_t = 1, \\ p_{t,h}[\text{mate}_k(y_t)], & r_t = 0.5, \\ 1 - p_{t,h}(y_t) - p_{t,h}[\text{mate}_k(y_t)], & r_t = 0. \end{cases} \tag{11}$$

Evidence memory applies the discounted update $\Pi_t^+(h,k) \propto [\Pi_t^-(h,k)]^\gamma \ell_t(h,k)$, normalized across active rule–pairing combinations. Summing over pairings yields each rule's belief. This joint representation lets partial feedback refine classification as response-pair knowledge develops.

For precision updating, the pairing distribution is evaluated after memory discounting and before incorporating the current feedback:

$$\widetilde{q_t^-}(k \mid h) = \frac{[\Pi_t^-(h,k)]^\gamma}{\sum_{k'=1}^{3} [\Pi_t^-(h,k')]^\gamma}. \tag{12}$$

When $\gamma = 0$, active rule–pairing combinations receive equal weight. A rule with zero total joint belief uses a uniform conditional pairing distribution.

The pairing-averaged feedback likelihood is $\ell_t(h) = \sum_k \widetilde{q_t^-}(k \mid h)\ell_t(h,k)$. This likelihood enters the precision gradient and update in Eqs. (6) and (7). Differentiation varies precision in the category predictions while holding the discounted pairing distribution and category distances fixed.

During candidate changes, retained rules preserve their conditional pairing beliefs. An entrant inherits the overall pairing distribution obtained after feedback and before candidate removal, carrying learned family relations into the new hypothesis.

Inference from choices

Particle filtering represents uncertainty over possible internal learning trajectories. Each particle n carries a candidate set and its associated cognitive states. Its normalized weight $W_t^{(n),\mathrm{pre}}$ expresses the researcher's support for that trajectory from preceding observations. The predictive choice probability and weights after observing the current choice are

$$\widehat{P}_t(c) = \sum_n W_t^{(n),\mathrm{pre}} P_t(c \mid z_t^{(n)}), \tag{13}$$

$$W_t^{(n),\mathrm{choice}} = \frac{W_t^{(n),\mathrm{pre}} P_t(y_t \mid z_t^{(n)})}{\widehat{P}_t(y_t)}. \tag{14}$$

Each particle then updates the learner's states using the observed feedback. Within a trajectory, feedback changes the learner's beliefs about rules. Across trajectories, choices change the researcher's support for possible internal states. The likelihood thus averages over uncertainty about perception and candidate dynamics.

The full model defines 13 participant-level parameters, including β_0 and η for precision learning. Free and fixed parameters are specified before estimation. Participant parameters $\boldsymbol{\theta}$ are estimated by minimizing mean choice negative log-likelihood,

$$\mathrm{NLL}(\boldsymbol{\theta}) = -\frac{1}{T} \sum_{t \in \mathcal{T}_{\mathrm{score}}} \log \widehat{P}_t(y_t; \boldsymbol{\theta}), \tag{15}$$

where $\mathcal{T}_{\mathrm{score}}$ is the set of scored trials and T its size. Predictions are formed before current feedback. Choice likelihood is the fitting objective.

[Implementation and validation of the revised precision update remain pending. Analysis cohort, estimation settings, training and evaluation boundaries, and uncertainty procedures to be specified.]

Interpreting reconstructed trajectories

Weighted particle states describe when a rule becomes available, how much belief it receives, how strongly it influences choice and how concentrated its predictions become. These quantities distinguish hypothesis discovery, evidence accumulation and policy expression within individual learning sequences. Candidate-set size measures the number of available explanations; choice-weight concentration measures their effective participation. Their timing can be examined jointly to identify delays between availability, support and dominance. Pre-choice weights summarize states using information available before the current response, and choice-conditioned weights incorporate that response. Analyses specify which conditioning is used and retain the distribution across trajectories as the model's uncertainty about the learner's internal state.

Independent strategy validation

Verbal reports are reserved for independent evaluation of reconstructed strategies. They are excluded from parameter fitting and particle inference. Validation compares model-predicted strategy summaries with reports encoded according to a prespecified scheme. Comparisons account for the timing of each report and the uncertainty across inferred trajectories.

[Report encoding, model-to-report alignment and validation statistics to be specified.]